

Answer key for the uploaded placement test:  PlacementAssessmentTest1-1

1. $-1 < w < 1$; graph open circles at -1 and 1 , shaded between.
2. (a) 4 , (b) -3 , (c) $-\frac{1}{3}$
3. $6x^3 - 15x^2 + 15x - 9$
4. $\frac{x-9}{x-5}$
5. NO REAL SOLUTION
6. $(-\infty, -\frac{5}{2}] \cup [0, \infty)$
7. $y = \frac{6}{7}x + 5$
8. $f(z) = 5 - 4z^2$. Net change: -16 . Average rate of change: -8 .
9. $g(x) = x^2 - x - 20$. Best viewing window: $[-7, 7]$ by $[-25, 20]$. It opens upward with zeros -4 and 5 , vertex near $(0.5, -20.25)$.
10. (a) x^2 : IV. (b) x^3 : II. (c) $\sqrt{x}$: I. (d) $|x|$: III.
11. $P(x) = x(x + 5)(x - 4)$. Zeros: $-5, 0, 4$. Correct graph is the first graph shown.
12. (a) \$1215.31, (b) \$1476.98, (c) \$3221.99
13. $\sin(t) = -\frac{4}{5}$, $\cos(t) = -\frac{3}{5}$, $\tan(t) = \frac{4}{3}$
14. $x = 34\sqrt{2}$
15. $\frac{\cos(u) \sec(u)}{\tan(u)} = \frac{\cos(u) \cdot \frac{1}{\cos(u)}}{\tan(u)} = \frac{1}{\tan(u)} = \cot(u)$
16. 14
17. $6(9) + 7(10) + 8(11) + 9(12)$, or $54 + 70 + 88 + 108$.